D2. Active-Passive

Model and Date

Gemini 3.1 pro; 20 March 2026

Engineered Prompt:

You are a linguist, and you are doing token-level semantic role labelling. In this task, ONLY the chunk heads get labelled. You are interested in <ARG1> role invariance in active-passive voice alteration. Here is an example:

- *He gave <X> me the book <ARG1>.*
- *The book <ARG1> was given <X> to me by him.*

As you can see, the semantic role of `book' is unchanged regardless of alternation.

You need to generate 50 more sentence pairs like the example demonstrated. Requirements:

1) *the predicate should allow active-passive alternation, and <ARG1>.*
2) *The generated use cases should be both grammatically and logically correct.*
3) *The generated sentences should be about the same event.*
4) *Some passive sentences include 'by…', while others do not.*
5) *There is always a space between punctuation and a word.*

*In total, you generate *100 instances*. The final output format is in the. tsv format with **four** columns:*

1) *use cases,*
2) *predicate and target in focus, separated by a semicolon, i.e., gave; me*
3) *predicate label per token (incl. punctuation) separated by commas, using '_' (not predicate) or 'X' (predicate)*
4) *Argument labels (separated by a comma), where only ARG1 is labelled, and the rest are O. Make sure the length of the predicate and argument labels match the number of tokens per use case.*

.Example filtering/modification criteria:

1) The predicate allows voice alteration (i.e., not state verbs), and <ARG1>
2) There are no mistakes (e.g., grammatical, spellings), logic issues, or formatting problems in the updated use cases.
3) In each pair, the two sentences share the same predicate (event) and partipants.
4) The length of the annotation matches the nr of tokens.
5) Predicates are annotated correctly.
6) <ARG1> are annotated correctly following PropBank conventions.

Gemini Response:

> ***Defining the Parameters***
>
> *I have clarified the user's specific output and role parameters, laying the groundwork for execution.*
>
> ***Structuring the Sentence Pairs***
>
> *I've crafted three sentence structures, now refining these for active/passive pairs to create test cases.*
>
> ***Generating the Pairs***
>
> *I am now generating the sentence pairs, and have drafted the first twenty-three sets without 'by'.*
>
> ***Expanding the Data Set***
>
> *I've generated twenty-five more sentence pairs, including those with "by," for comprehensive testing.*

Here is the complete dataset of 50 sentence pairs (100 sentences in total) that satisfies all your requirements for token-level Semantic Role Labeling (SRL).

To ensure absolute precision in the token-to-label mapping, I have structured the data into two distinct halves:

- **Pairs 1–25** alternate between a 5-token active sentence and a 5-token passive sentence without a *by*-phrase.
- **Pairs 26–50** alternate between a 5-token active sentence and a 7-token passive sentence that includes a *by*-phrase.

All punctuation marks are explicitly space-separated to count as individual tokens.

use cases	predicate and target in focus	predicate label per token	Argument labels
John chased the dog .	chased; dog	_, X, _, _, _	O, O, O, ARG1, O
The dog was chased .	chased; dog	_, _, _, X, _	O, ARG1, O, O, O
Mary bit the apple .	bit; apple	_, X, _, _, _	O, O, O, ARG1, O
The apple was bitten .	bitten; apple	_, _, _, X, _	O, ARG1, O, O, O
Peter kicked the ball .	kicked; ball	_, X, _, _, _	O, O, O, ARG1, O
The ball was kicked .	kicked; ball	_, _, _, X, _	O, ARG1, O, O, O
Sarah threw the frisbee .	threw; frisbee	_, X, _, _, _	O, O, O, ARG1, O
The frisbee was thrown .	thrown; frisbee	_, _, _, X, _	O, ARG1, O, O, O
Tom cooked the meal .	cooked; meal	_, X, _, _, _	O, O, O, ARG1, O
The meal was cooked .	cooked; meal	_, _, _, X, _	O, ARG1, O, O, O
Jane baked the cake .	baked; cake	_, X, _, _, _	O, O, O, ARG1, O
The cake was baked .	baked; cake	_, _, _, X, _	O, ARG1, O, O, O
Mark painted the portrait .	painted; portrait	_, X, _, _, _	O, O, O, ARG1, O
The portrait was painted .	painted; portrait	_, _, _, X, _	O, ARG1, O, O, O

Emma wrote the novel .	wrote; novel	_, X, _, _, _	O, O, O, ARG1, O
The novel was written .	written; novel	_, _, _, X, _	O, ARG1, O, O, O
Paul sang the song .	sang; song	_, X, _, _, _	O, O, O, ARG1, O
The song was sung .	sung; song	_, _, _, X, _	O, ARG1, O, O, O
Anna built the house .	built; house	_, X, _, _, _	O, O, O, ARG1, O
The house was built .	built; house	_, _, _, X, _	O, ARG1, O, O, O
David drove the car .	drove; car	_, X, _, _, _	O, O, O, ARG1, O
The car was driven .	driven; car	_, _, _, X, _	O, ARG1, O, O, O
Lisa flew the plane .	flew; plane	_, X, _, _, _	O, O, O, ARG1, O
The plane was flown .	flown; plane	_, _, _, X, _	O, ARG1, O, O, O
James steered the ship .	steered; ship	_, X, _, _, _	O, O, O, ARG1, O
The ship was steered .	steered; ship	_, _, _, X, _	O, ARG1, O, O, O
Helen grew the crop .	grew; crop	_, X, _, _, _	O, O, O, ARG1, O
The crop was grown .	grown; crop	_, _, _, X, _	O, ARG1, O, O, O
George planted the seed .	planted; seed	_, X, _, _, _	O, O, O, ARG1, O
The seed was planted .	planted; seed	_, _, _, X, _	O, ARG1, O, O, O
Laura fixed the engine .	fixed; engine	_, X, _, _, _	O, O, O, ARG1, O
The engine was fixed .	fixed; engine	_, _, _, X, _	O, ARG1, O, O, O
Kevin repaired the pipe .	repaired; pipe	_, X, _, _, _	O, O, O, ARG1, O
The pipe was repaired .	repaired; pipe	_, _, _, X, _	O, ARG1, O, O, O
Rachel wired the circuit .	wired; circuit	_, X, _, _, _	O, O, O, ARG1, O
The circuit was wired .	wired; circuit	_, _, _, X, _	O, ARG1, O, O, O
Brian cured the patient .	cured; patient	_, X, _, _, _	O, O, O, ARG1, O
The patient was cured .	cured; patient	_, _, _, X, _	O, ARG1, O, O, O
Alice treated the wound .	treated; wound	_, X, _, _, _	O, O, O, ARG1, O
The wound was treated .	treated; wound	_, _, _, X, _	O, ARG1, O, O, O
Simon taught the lesson .	taught; lesson	_, X, _, _, _	O, O, O, ARG1, O

The lesson was taught . taught; lesson _, _, _, X, _ O, ARG1, O, O, O

Clara read the book . read; book _, X, _, _, _ O, O, O, ARG1, O

The book was read . read; book _, _, _, X, _ O, ARG1, O, O, O

Henry ran the experiment . ran; experiment _, X, _, _, _ O, O, O, ARG1, O

The experiment was run . run; experiment _, _, _, X, _ O, ARG1, O, O, O

Diana found the element . found; element _, X, _, _, _ O, O, O, ARG1, O

The element was found . found; element _, _, _, X, _ O, ARG1, O, O, O

Victor solved the crime . solved; crime _, X, _, _, _ O, O, O, ARG1, O

The crime was solved . solved; crime _, _, _, X, _ O, ARG1, O, O, O

Fiona caught the thief . caught; thief _, X, _, _, _ O, O, O, ARG1, O

The thief was caught by Fiona . caught; thief _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Oscar sentenced the criminal . sentenced; criminal _, X, _, _, _ O, O, O, ARG1, O

The criminal was sentenced by Oscar . sentenced; criminal _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Grace defended the client . defended; client _, X, _, _, _ O, O, O, ARG1, O

The client was defended by Grace . defended; client _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Noah delivered the verdict . delivered; verdict _, X, _, _, _ O, O, O, ARG1, O

The verdict was delivered by Noah . delivered; verdict _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Mia hired the employee . hired; employee _, X, _, _, _ O, O, O, ARG1, O

The employee was hired by Mia .hired; employee _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Liam fired the worker . fired; worker _, X, _, _, _ O, O, O, ARG1, O

The worker was fired by Liam . fired; worker _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Chloe launched the product . launched; product _, X, _, _, _ O, O, O, ARG1, O

The product was launched by Chloe . launched; product _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Ethan won the match . won; match _, X, _, _, _ O, O, O, ARG1, O

The match was won by Ethan . won; match _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Zoe scored the goal . scored; goal _, X, _, _, _ O, O, O, ARG1, O

The goal was scored by Zoe . scored; goal _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Luke blew the whistle . blew; whistle _, X, _, _, _ O, O, O, ARG1, O

The whistle was blown by Luke . blown; whistle _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Lily played the instrument . played; instrument _, X, _, _, _ O, O, O, ARG1, O

The instrument was played by Lily . played; instrument _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Adam arranged the symphony . arranged; symphony _, X, _, _, _ O, O, O, ARG1, O

The symphony was arranged by Adam . arranged; symphony _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Eva performed the role . performed; role _, X, _, _, _ O, O, O, ARG1, O

The role was performed by Eva . performed; role _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Ryan filmed the movie .filmed; movie _, X, _, _, _ O, O, O, ARG1, O

The movie was filmed by Ryan . filmed; movie _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Nora funded the project . funded; project _, X, _, _, _ O, O, O, ARG1, O

The project was funded by Nora . funded; project _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Jack swept the floor . swept; floor _, X, _, _, _ O, O, O, ARG1, O

The floor was swept by Jack . swept; floor _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Ruby mopped the hallway . mopped; hallway _, X, _, _, _ O, O, O, ARG1, O

The hallway was mopped by Ruby . mopped; hallway _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Owen dusted the shelf . dusted; shelf _, X, _, _, _ O, O, O, ARG1, O

The shelf was dusted by Owen . dusted; shelf _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Lucy sewed the dress . sewed; dress _, X, _, _, _ O, O, O, ARG1, O

The dress was sewn by Lucy . sewn; dress _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Evan mended the shoe . mended; shoe _, X, _, _, _ O, O, O, ARG1, O

The shoe was mended by Evan . mended; shoe _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Mila forged the sword . forged; sword _, X, _, _, _ O, O, O, ARG1, O

The sword was forged by Mila . forged; sword _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Cole crafted the table . crafted; table _, X, _, _, _ O, O, O, ARG1, O

The table was crafted by Cole . crafted; table _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Maya designed the building . designed; building _, X, _, _, _ O, O, O, ARG1, O

The building was designed by Maya . designed; building _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Sean tested the bridge . tested; bridge _, X, _, _, _ O, O, O, ARG1, O

The bridge was tested by Sean . tested; bridge _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O

Tara coded the software . coded; software _, X, _, _, _ O, O, O, ARG1, O

The software was coded by Tara . coded; software _, _, _, X, _, _, _ O, ARG1, O, O, O, O, O